

Software Cost Estimation

Function Point Analysis & Use Case Points

Traffic Prediction and Analysis System

Assignment No. 8

Group: E-16

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1. Problem Statement

Perform software cost estimation exercises using **Function Point Analysis (FPA)** and **Use Case Points (UCP)** for the Traffic Prediction and Analysis System. Estimate the software size, effort, and approximate development cost based on these two industry-standard techniques.

2. Introduction

Software cost estimation is a fundamental activity in project management that predicts the resources, time, and budget required to develop a software system. Accurate estimation helps prevent budget overruns, enables proper resource allocation, and sets realistic delivery expectations.

Two widely used estimation techniques are:

- i) **Function Point Analysis (FPA)** — measures software size based on the functional requirements delivered to the user, independent of the programming language or technology used.
- ii) **Use Case Points (UCP)** — estimates effort based on the complexity of use cases and the environmental and technical factors of the project.

This assignment applies both techniques to the **Traffic Prediction and Analysis System**, a full-stack web application built using React.js, Node.js/Express, and MongoDB that integrates external APIs and a MobileNet ML model for traffic forecasting and pothole detection.

3. Function Point Analysis (FPA)

3.1 Overview

Function Point Analysis, introduced by Allan Albrecht at IBM in 1979, quantifies software functionality from the user's perspective. It classifies system interactions into five categories:

- **External Inputs (EI)** — data entering the system from outside.
- **External Outputs (EO)** — data or reports leaving the system.
- **External Inquiries (EQ)** — input/output combinations for data retrieval.
- **Internal Logical Files (ILF)** — data maintained within the system.
- **External Interface Files (EIF)** — data referenced from external systems.

3.2 Step 1: Identify and Classify Function Points

External Inputs (EI)

#	Function	Complexity
1	User Registration	Low
2	User Login (JWT Authentication)	Low
3	Submit Traffic Prediction Request	Average
4	Upload Pothole Image	High
5	Submit Infrastructure Complaint	Average
6	Register Event Space	Low
7	Add Construction Data (Admin)	Average
8	Add Traffic Diversion (Admin)	Average
9	Update User Role (Admin)	Low

External Outputs (EO)

#	Function	Complexity
1	Traffic Prediction Result (Score + Breakdown)	High
2	Time Slot Suggestions	Average
3	Citizen Score Display	Low
4	Admin Analytics Dashboard	High
5	Complaint Status Report	Average

External Inquiries (EQ)

#	Function	Complexity
1	View Real-time Traffic Map	Average
2	Search Route Information	Average
3	View Complaint History	Low
4	View Pothole Reports	Low
5	View User List (Admin)	Low

Internal Logical Files (ILF)

#	Data Store	Complexity
1	Users Collection	Low
2	PathInfo (Traffic Scores)	High
3	Complaints Collection	Average
4	Images (Pothole Reports)	Average
5	Events Collection	Low
6	Construction Collection	Low
7	Diversions Collection	Low
8	HotspotLocations Collection	Average
9	PredictionLogs Collection	Average
10	TrafficStatus Collection	Average

External Interface Files (EIF)

#	External System	Complexity
1	Google Maps Directions API	High
2	OpenWeatherMap API	Average
3	Calendarific Events API	Low
4	MobileNet ML Model	High
5	Cloudinary Image Storage	Average

3.3 Step 2: Calculate Unadjusted Function Points (UFP)

The weighting factors for each complexity level are:

Type	Low	Average	High
External Inputs (EI)	3	4	6
External Outputs (EO)	4	5	7
External Inquiries (EQ)	3	4	6
Internal Logical Files (ILF)	7	10	15
External Interface Files (EIF)	5	7	10

Calculation:

Type	Low	Average	High	Total
EI	$4 \times 3 = 12$	$4 \times 4 = 16$	$1 \times 6 = 6$	34
EO	$1 \times 4 = 4$	$2 \times 5 = 10$	$2 \times 7 = 14$	28
EQ	$3 \times 3 = 9$	$2 \times 4 = 8$	—	17
ILF	$4 \times 7 = 28$	$5 \times 10 = 50$	$1 \times 15 = 15$	93
EIF	$1 \times 5 = 5$	$2 \times 7 = 14$	$2 \times 10 = 20$	39
Unadjusted Function Points (UFP)				211

3.4 Step 3: Value Adjustment Factor (VAF)

The VAF is calculated using 14 General System Characteristics (GSCs), each rated on a scale of 0–5.

#	Characteristic	Rating (0–5)
1	Data Communications	5
2	Distributed Data Processing	3
3	Performance	4
4	Heavily Used Configuration	2
5	Transaction Rate	3
6	Online Data Entry	5
7	End-User Efficiency	4
8	Online Update	4
9	Complex Processing	4
10	Reusability	3
11	Installation Ease	3
12	Operational Ease	3
13	Multiple Sites	4
14	Facilitate Change	3
Total Degree of Influence (TDI)		50

$$\text{VAF} = 0.65 + (0.01 \times \text{TDI}) = 0.65 + (0.01 \times 50) = \mathbf{1.15}$$

3.5 Step 4: Adjusted Function Points (AFP)

$$\text{AFP} = \text{UFP} \times \text{VAF} = 211 \times 1.15 = \mathbf{242.65 \approx 243 \text{ FP}}$$

3.6 Step 5: Effort and Cost Estimation

Using industry averages for a web application project:

- Productivity rate: **10 FP per person-month** (for a MERN stack project of moderate complexity)
- Average developer cost: **INR 40,000 per month** (academic/intern-level team)

$$\text{Effort} = \frac{\text{AFP}}{\text{Productivity}} = \frac{243}{10} = \mathbf{24.3 \text{ person-months}}$$

With a team of 4 developers:

$$\text{Duration} = \frac{24.3}{4} \approx \mathbf{6.1 \text{ months}}$$

$$\text{Total Cost} = 24.3 \times 40,000 = \text{INR } \mathbf{9,72,000}$$

4. Use Case Points (UCP) Estimation

4.1 Overview

The Use Case Points method, proposed by Gustav Karner in 1993, estimates effort based on the complexity of actors and use cases, adjusted by technical and environmental factors.

4.2 Step 1: Unadjusted Actor Weight (UAW)

Actors are classified by their interaction type:

Actor	Type	Weight	Count
Citizen User	Complex (GUI-based)	3	1
Municipal Admin	Complex (GUI-based)	3	1
Google Maps API	Simple (API)	1	1
OpenWeatherMap API	Simple (API)	1	1
Calendarific API	Simple (API)	1	1
MobileNet ML Model	Average (protocol-driven)	2	1
Cloudinary	Simple (API)	1	1
		UAW	12

4.3 Step 2: Unadjusted Use Case Weight (UUCW)

Use cases are classified by transaction count:

Use Case	Complexity	Weight	Transactions
Register Account	Simple (≤ 3)	5	3
Login / Logout	Simple	5	2
View Traffic Prediction	Complex (> 7)	15	12
Get Time Suggestions	Average (4–7)	10	5
Upload Pothole Image	Complex	15	8
Submit Complaint	Average	10	5
Register Event Space	Simple	5	3
View Citizen Score	Simple	5	2
View Traffic Map	Average	10	6
Add Construction Data	Simple	5	3
Add Traffic Diversion	Simple	5	3
Review Citizen Reports	Average	10	5
Manage Users	Average	10	6
View Analytics Dashboard	Complex	15	10
		UUCW	125

4.4 Step 3: Unadjusted Use Case Points (UUCP)

$$\text{UUCP} = \text{UAW} + \text{UUCW} = 12 + 125 = \mathbf{137}$$

4.5 Step 4: Technical Complexity Factor (TCF)

T#	Factor	Wt.	Rating (0–5)
T1	Distributed System	2	4
T2	Response Time / Performance	1	4
T3	End-User Efficiency	1	4
T4	Complex Internal Processing	1	5
T5	Reusability	1	3
T6	Easy to Install	0.5	3
T7	Easy to Use	0.5	4
T8	Portability	2	4
T9	Easy to Change	1	3
T10	Concurrency	1	3
T11	Security Features	1	4
T12	Third-Party Access	1	5
T13	Training Required	1	2
TFactor = $\sum(\text{Wt.} \times \text{Rating})$			52.5

$$\text{TCF} = 0.6 + (0.01 \times 52.5) = \mathbf{1.125}$$

4.6 Step 5: Environmental Complexity Factor (ECF)

E#	Factor	Wt.	Rating (0–5)
E1	Familiarity with Project	1.5	3
E2	Application Experience	0.5	3
E3	OOP Experience	1	4
E4	Lead Analyst Capability	0.5	3
E5	Motivation	1	4
E6	Stable Requirements	2	3
E7	Part-Time Workers	−1	3
E8	Difficult Programming Language	−1	1
EFactor = $\sum(\text{Wt.} \times \text{Rating})$			17.5

$$\text{ECF} = 1.4 + (-0.03 \times 17.5) = 1.4 - 0.525 = \mathbf{0.875}$$

4.7 Step 6: Adjusted Use Case Points (UCP)

$$\text{UCP} = \text{UUCP} \times \text{TCF} \times \text{ECF} = 137 \times 1.125 \times 0.875 = \mathbf{134.86} \approx \mathbf{135 \text{ UCP}}$$

4.8 Step 7: Effort and Cost Estimation

Using the standard productivity factor of **20 person-hours per UCP**:

$$\text{Effort (hours)} = 134.86 \times 20 = \mathbf{2,697.2 \text{ person-hours}}$$

Converting to person-months (assuming 160 working hours/month):

$$\text{Effort (months)} = \frac{2697.2}{160} = \mathbf{16.86} \text{ person-months}$$

With a team of 4 developers:

$$\text{Duration} = \frac{16.86}{4} \approx \mathbf{4.2} \text{ months}$$

$$\text{Total Cost} = 16.86 \times 40,000 = \text{INR } \mathbf{6,74,400}$$

5. Comparison of Results

Parameter	FPA	UCP
Software Size	243 FP	135 UCP
Effort	24.3 person-months	16.9 person-months
Duration (4 developers)	6.1 months	4.2 months
Estimated Cost	INR 9,72,000	INR 6,76,000

The FPA method yields a higher estimate because it accounts for all internal data stores (10 MongoDB collections) and external interface files (5 APIs), which contribute significantly to the function point count. The UCP method, being use-case driven, focuses on user-facing interactions and tends to produce a more conservative estimate for backend-heavy systems.

The realistic estimate for this project lies between the two values, approximately **20 person-months** of total effort, translating to roughly **5 months** of development with a 4-member team.

6. Conclusion

This assignment demonstrated two complementary approaches to software cost estimation for the Traffic Prediction and Analysis System. Function Point Analysis provided a comprehensive size measurement accounting for all data interactions and system boundaries, yielding 243 adjusted function points. The Use Case Points method offered an effort-centric estimate grounded in use case complexity and team environment factors, producing 129 adjusted use case points.

Together, these techniques provide upper and lower bounds for project planning, enabling the development team to set realistic milestones, allocate resources proportionally, and manage stakeholder expectations with data-driven estimates.